Infinite series: answers

Math 163 • Prof. Chowdhury • Concept map

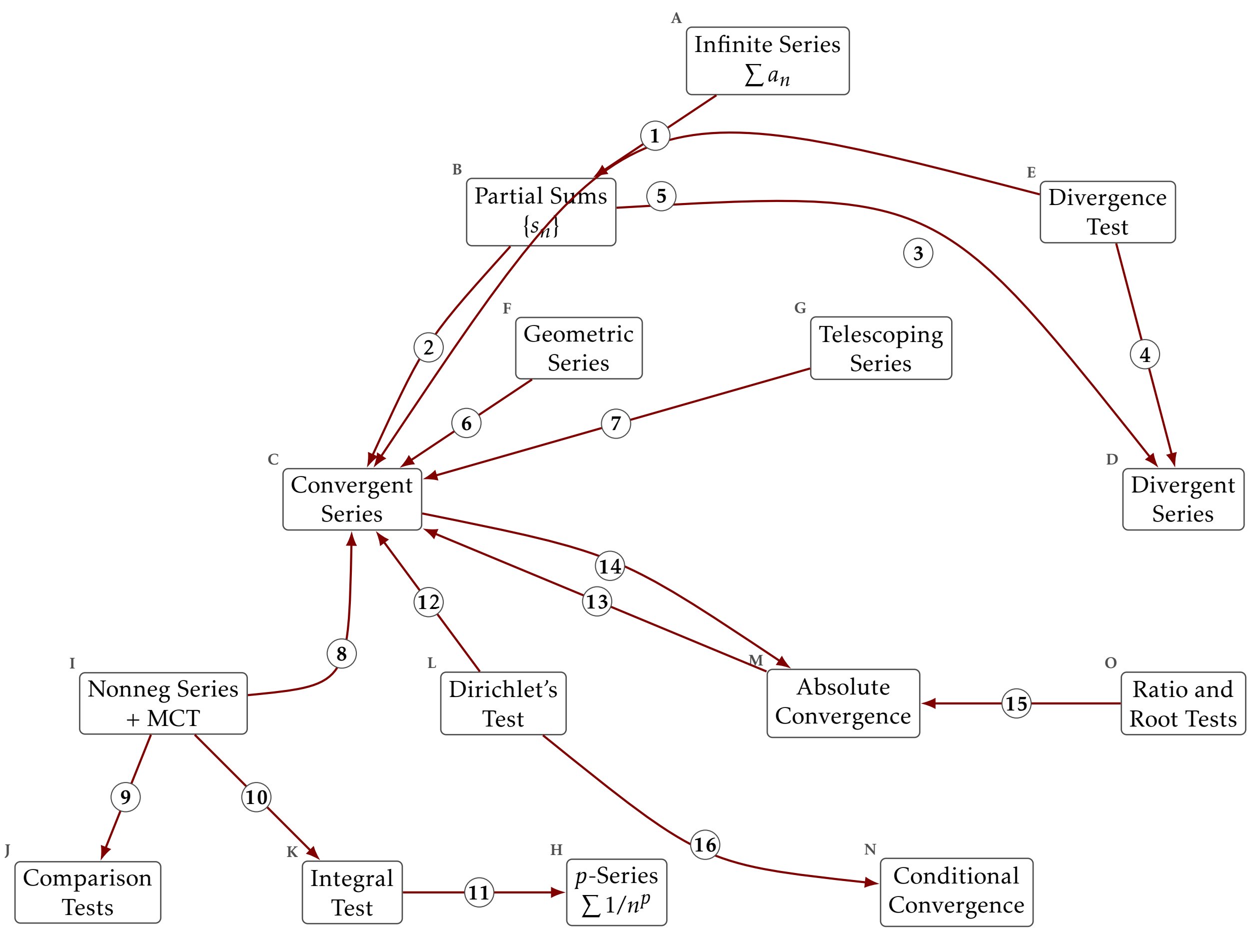
Every arrow named, with the kind of relationship it is. The arrows are drawn here in one style, as they are on the blank worksheet and on the web version.

What each box means

- **A. Infinite Series $\sum a_n$.** The expression $a_1 + a_2 + a_3 + \cdots$. It has no meaning until the partial sums give it one.
- **B. Partial Sums $\{s_n\}$.** $s_n = a_1 + \cdots + a_n$. An ordinary sequence, so everything you know about sequences applies to it.
- **C. Convergent Series.** $\sum a_n$ converges when $\{s_n\}$ has a finite limit, and the sum is that limit.
- **D. Divergent Series.** $\sum a_n$ diverges when $\{s_n\}$ fails to converge, whether it runs away or oscillates.
- **E. Divergence Test.** If $a_n \not\to 0$ then $\sum a_n$ diverges. It is a test for divergence only, and it never certifies convergence.
- **F. Geometric Series.** $\sum ar^n$, which converges exactly when $|r| < 1$, with sum $a/(1-r)$ and partial sums $a(1-r^n)/(1-r)$.
- **G. Telescoping Series.** $\sum(b_n - b_{n+1})$, whose partial sums collapse to $b_1 - b_{n+1}$.
- **H. p-Series $\sum 1/n^p$.** Converges exactly when $p > 1$. The boundary $p = 1$ is the harmonic series, which diverges.
- **I. Nonneg Series + MCT.** If $a_n \geq 0$ the partial sums increase, so the monotone convergence theorem turns convergence into boundedness.
- **J. Comparison Tests.** Direct: $0 \leq a_n \leq cb_n$ eventually and $\sum b_n$ convergent gives $\sum a_n$ convergent. Limit: $\lim a_n/b_n \in (0,\infty)$ gives the same behaviour for both.
- **K. Integral Test.** If f is positive, continuous and eventually decreasing with $f(n) = a_n$, then $\sum a_n$ and $\int_1^\infty f$ converge or diverge together.
- **L. Dirichlet's Test.** If a_n decreases to 0 and the partial sums of b_n are bounded, then $\sum a_n b_n$ converges. Proved by summation by parts.
- **M. Absolute Convergence.** $\sum a_n$ converges absolutely when $\sum |a_n|$ converges. This is strictly stronger than convergence.
- **N. Conditional Convergence.** $\sum a_n$ converges conditionally when it converges and $\sum |a_n|$ does not. The convergence rests on cancellation.
- **O. Ratio and Root Tests.** With $L = \lim |a_{n+1}/a_n|$ or $L = \limsup |a_n|^{1/n}$: absolutely convergent if $L1$, and no conclusion if $L = 1$.

Examples worth keeping to hand

- The harmonic series $\sum 1/n$ diverges even though its terms tend to zero.
- The alternating harmonic series $\sum(-1)^{n+1}/n$ converges conditionally, to $\ln 2$.
- $\sum \sin(n\theta)/n$ converges by Dirichlet's test and is not alternating.
- Summation by parts is the discrete integration by parts, and it is what proves Dirichlet's test.
- The Cauchy criterion for series: control every far enough tail at once, not one chosen tail.


A
Infinite Series $\sum a_n$
B
Partial Sums $\{s_n\}$
E
Divergence Test
C
Convergent Series
D
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F
Geometric Series
G
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J
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K
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p-Series $\sum 1/n^p$
N
Conditional Convergence
1
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The arrows

1. **holds.** Convergence of the series is defined to be convergence of its sequence of partial sums.
2. **holds.** If the partial sums have a finite limit then the series converges, and its sum is that limit.
3. **holds.** If the partial sums fail to converge then the series diverges, and running off to infinity is only one of the ways that happens.
4. **holds.** Terms that do not tend to zero leave the partial sums unable to settle, so the series diverges.
5. **fails.** Terms tending to zero is necessary for convergence and not sufficient for it, and the harmonic series is the witness.
6. **holds.** A geometric series converges exactly when its ratio has modulus below one, and then the closed form for the partial sums gives the sum.
7. **holds.** In a telescoping series consecutive terms cancel, so the partial sums collapse and converge exactly when b_n does.
8. **holds.** For a series of nonnegative terms the partial sums increase, so the series converges exactly when they are bounded above.
9. **holds.** Bounding a nonnegative series term by term against a series you already know bounds its partial sums.
10. **holds.** Comparing a decreasing positive term with the area under the matching function bounds the partial sums by an improper integral.
11. **holds.** Running the integral test on $1/x^p$ puts the threshold for the p-series at $p = 1$.
12. **holds.** A sequence decreasing to zero paired with bounded partial sums gives a convergent series, and the alternating series test is the case where the second factor is $(-1)^n$.
13. **holds.** A series whose terms are absolutely summable converges, so the absolute version is the stronger property.
14. **fails.** A convergent series need not converge absolutely, and the alternating harmonic series is the witness.
15. **holds.** The ratio and root tests compare the sizes of the terms against a geometric series, so what they establish is the absolute version.
16. **holds.** A test that gives convergence without touching the absolute version can certify a series whose absolute version diverges.

One last question

Which single arrow carries the most of this chapter? Say why in one sentence. There is more than one defensible answer, and the argument is the useful part.